MUHAMMAD ASAD LAL



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ABOUT ME

I am currently in the final semester of my Master's in Computer Science and Technology at Nanjing University of Science and Technology. My research focuses on evolutionary optimization, particularly adaptive and structure-aware search. I am interested in extending this work toward learning-assisted optimization and algorithm design, where structural information and search experience can guide more efficient optimization decisions.

RESEARCH DIRECTIONS

- Evolutionary & Metaheuristic Optimization
- Learning-Assisted Optimization
- Adaptive & Structure-Aware Search
- Machine Learning for Optimization

PUBLICATIONS *Google Scholar*

1. **Lal, M.A.**, Kashif, M. & Huang, B. Father-inspired metaheuristic: a hybrid framework with neighborhood repulsion and elite learning for continuous optimization. *Evol. Intel. 19, 111 (2026)*.
   DOI: 10.1007/s12065-026-01228-w
2. **Lal, M.A.**, Kashif, M., Lv, J. & Huang, B. Multilevel Otsu Thresholding of Fundus Images Using Augmented Forman-Ricci Curvature-Guided SHADE. Biomed. Signal Process. Control. *Under Review (2026)*.
3. Kashif, M., **Lal, M.A.** & Huang, B. Learn-to-Adapt: A Hybrid Optimizer with Bidirectional Elite Guidance and Gradient-Assisted Refinement. *Swarm Evol. Comput. Under Review (2026)*.
4. Kashif, M., **Lal, M.A.** & Huang, B. ISHADE-DPM: An Improved SHADE with Dynamic Parameter Memory for Efficient Continuous Optimization. *Evol. Comput. Submitted (2026)*.

ACADEMIC QUALIFICATION

Master (Computer Science & Technology) **Sep 2024 – Apr 2027**
Nanjing University of Science and Technology, China

Bachelor (Computer Science), CGPA 3.82/4.00 (Top in Class) **Aug 2019 – Sep 2023**
Government College University Faisalabad, Pakistan

RESEARCH PROJECTS

RAVO: Region-Adaptive Variable-Space Optimizer **Sep 2026 – Present**

- Developing a region-adaptive optimizer with productivity-based FE allocation.
- Using region-conditioned donor roles for guided and diversity-aware search.
- Adapting crossover between original and decorrelated variable spaces.

AFRC-SHADE: Augmented Forman-Ricci Curvature-Guided SHADE **Jan 2026 – Aug 2026**

- Developed a structure-aware SHADE variant using graph-based population information.
- Used curvature-guided indicators to adapt mutation strength and search behavior.
- Applied AFRC-SHADE to multi-level threshold image segmentation.

UNDERGRADUATE PROJECTS

Heart Disease Prediction (Final Year Project) **Aug 2019 – Sep 2023**

- Developed an Android app for heart disease prediction.
- Used Logistic Regression for health classification.

PROFESSIONAL EXPERIENCE

Android Development (Internship) **Oct 2023 – Mar 2024**
Idealogic Software House

- Contributed to Android app development projects with the team.
- Improved practical skills in Android development, programming, and software practices.

PROFESSIONAL AND PERSONAL SKILLS

- MATLAB
- C/C++
- LaTex
- Python
- VS Code
- GitHub

FELLOWSHIPS AND AWARDS

Fully Funded High Achievers (Lanshan) Scholarship **Sep. 01, 2024**
Nanjing University of Science and Technology, China

Outstanding Academic Performance Award **Sep. 29, 2023**
Nanjing University of Science and Technology, China

CERTIFICATIONS

One Million Prompters **Jun. 24, 2026**
Dubai Future Foundation

WIPO – China (NJUST) Summer School on Intellectual Property **Jul. 11, 2025**
Nanjing University of Science and Technology

Chinese Writing Competition **Dec. 31, 2024**
Nanjing University of Science and Technology

Manage Sensitive Data and Security Policies within GitHub **Apr. 01, 2024**
Microsoft

Machine Learning with Python **Nov. 22, 2023**
International Business Machines (IBM)

Deep Learning with TensorFlow, Certification **Nov. 06, 2023**
International Business Machines (IBM)

Computer Application and Office Professional, Certification **Oct. 17, 2017**
Government of Punjab Vocational Training Council

EXTRA-CURRICULAR ACTIVITIES

International Student Registration Volunteer **Sep. 2025-26**
Nanjing University of Science and Technology, China

Xuanwu Youth Development Activity **Mar. 17, 2026**
United Nations Department of Safety and Security

BSAFE Security Awareness Training **May. 26, 2022**
United Nations Department of Safety and Security

REFERENCE

References will be provided upon request.